Test Patch Masonry — Masonry, Plaster & Finishes BOQ Report

1. Executive Summary & Code Compliance

Item / Parameter	Value	Standard Reference
Building Footprint	10.00 m × 12.00 m	Architectural Layout
Total Stories / Height	2 Stories / 6.0 m	NBC 2016 Part 3
Total Clay Bricks	6,264 Nos	IS 1077:1992 (Modular 20×20×10 cm)
Total Mortar Cement	36 Bags (1,798 kg)	IS 269:2015 (1:4 Mix Proportion)
Total Mortar Sand	7.99 Tonnes	IS 383:2016 Zone II
Total Plaster Surface Area	217.8 m²	IS 1200 Part 12
Total Plaster Cement	20 Bags (983 kg)	IS 1661:1972 (12mm int, 15mm ext)
Total Plaster Sand	5.16 Tonnes	IS 383:2016 Zone III
Estimated Wall & Finishes Cost	INR 222,156.00	CPWD DSR 2023

2. Masonry Brickwork Schedule & Material Takeoff (IS 1077, IS 2250)

Bricks modular 200×200×100 mm (250 Nos/m³). Mortar 1:4 mix proportion, dry factor 1.33, W:C = 0.50.

Wall ID	Fl	Span / Cols	Len (m)	Ht (m)	Thk (mm)	Gross Vol (m³)	Bricks (Nos)	Brick Vol (m³)	Wet Mortar (m³)	Dry Mortar (m³)	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W1_F0	F0	Bay	10.00	2.48	230	5.693	1,424	4.627	1.066	1.418	9	408.3	1.815	204
W2_F0	F0	Bay	12.00	2.48	230	6.831	1,708	5.549	1.282	1.705	10	490.9	2.182	245
W1_F1	F1	Bay	10.00	2.48	230	5.693	1,424	4.627	1.066	1.418	9	408.3	1.815	204
W2_F1	F1	Bay	12.00	2.48	230	6.831	1,708	5.549	1.282	1.705	10	490.9	2.182	245
TOTAL	**2Fl**	**All Walls**	**-**	**-**	**-**	**25.05**	**6,264**	**20.35**	**4.70**	**6.24**	**36**	**1,798**	**7.99**	**899**

3. Plastering Schedule & Quantities (IS 1661, IS 2402, IS 1200)

Ext: 15mm 1:4 render; Int: 12mm 1:6 plaster; Ceiling: 6mm 1:3 soffit plaster per IS 1661 Cl. 12.3. Dry factor 1.33.

Surface ID	Fl	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W1_EXT_F0	F0	Ext Render (15mm 1:4)	24.75	15	1:4	0.371	0.494	3	142.2	0.632	71
W1_INT_F0	F0	Int Wall (12mm 1:6)	24.75	12	1:6	0.297	0.395	2	81.3	0.542	41
W2_EXT_F0	F0	Ext Render (15mm 1:4)	29.70	15	1:4	0.446	0.593	4	170.6	0.758	85
W2_INT_F0	F0	Int Wall (12mm 1:6)	29.70	12	1:6	0.356	0.474	2	97.5	0.650	49
W1_EXT_F1	F1	Ext Render (15mm 1:4)	24.75	15	1:4	0.371	0.494	3	142.2	0.632	71
W1_INT_F1	F1	Int Wall (12mm 1:6)	24.75	12	1:6	0.297	0.395	2	81.3	0.542	41
W2_EXT_F1	F1	Ext Render (15mm 1:4)	29.70	15	1:4	0.446	0.593	4	170.6	0.758	85
W2_INT_F1	F1	Int Wall (12mm 1:6)	29.70	12	1:6	0.356	0.474	2	97.5	0.650	49
Ceil_F0	F0	Slab Soffit (6mm 1:3)	120.00	6	1:3	0.720	0.958	7	344.7	1.149	172
Ceil_F1	F1	Slab Soffit (6mm 1:3)	120.00	6	1:3	0.720	0.958	7	344.7	1.149	172
TOTAL	**2Fl**	**All Plaster Surfaces**	**457.8**	**-**	**-**	**4.380**	**5.826**	**34**	**1,673**	**7.46**	**836**

4. Painting & Surface Finishes Schedule (NBC 2016 SP7)

1 coat primer @ 8 m²/L, 2 coats polymer putty @ 1.5 kg/m², 2 coats emulsion (interior 12 m²/L, exterior 10 m²/L).

Surface ID	Fl	Location & Specification	Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Em 2-Ct (L)	Ext Guard 2-Ct (L)
W1_EXT_F0	F0	External Weather Guard	24.75	3.09	—	—	4.95
W1_INT_F0	F0	Int Acrylic Emulsion	24.75	3.09	74.25	4.12	—
W2_EXT_F0	F0	External Weather Guard	29.70	3.71	—	—	5.94
W2_INT_F0	F0	Int Acrylic Emulsion	29.70	3.71	89.10	4.95	—
W1_EXT_F1	F1	External Weather Guard	24.75	3.09	—	—	4.95
W1_INT_F1	F1	Int Acrylic Emulsion	24.75	3.09	74.25	4.12	—
W2_EXT_F1	F1	External Weather Guard	29.70	3.71	—	—	5.94
W2_INT_F1	F1	Int Acrylic Emulsion	29.70	3.71	89.10	4.95	—
Ceil_F0	F0	Slab Soffit (White Emulsion)	120.00	15.00	360.00	20.00	—
Ceil_F1	F1	Slab Soffit (White Emulsion)	120.00	15.00	360.00	20.00	—
TOTAL	**2Fl**	**All Surfaces (Walls + Ceilings)**	**457.8**	**57.2**	**1,046.7**	**58.1**	**21.8**

5. Consolidated Material Takeoff Summary (All Stories)

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Total Wall Gross Volume	L × t × Hv (clear storey height)	25.05	m³	IS 1905:1987
Clay Modular Building Bricks	Standard modular 20×20×10 cm	6,264	Nos	IS 1077:1992
Net Solid Brickwork Volume	Actual brick body vol. (excl. mortar)	20.35	m³	IS 1077:1992
Wet Brickwork Mortar Volume	Gross Wall Vol – Solid Brick Vol	4.70	m³	IS 2250:1981
Dry Brickwork Mortar Volume	Wet Volume × 1.33 dry factor	6.24	m³	IS 2250:1981
Brickwork Cement	1:4 Mix Proportion (OPC/PPC)	36 (1,798 kg)	Bags 50kg	IS 269:2015
Brickwork Sand	Coarse sand, dry density 1600 kg/m³	7.99	Tonnes	IS 383 Zone II
Brickwork Water	Water:Cement ratio = 0.50	899	Litres	IS 456 Cl 5.4
Wall Plaster Area	Int (12mm 1:6) + Ext (15mm 1:4)	217.8	m²	IS 1200 Pt 12
Ceiling Soffit Plaster Area	Soffit of slabs (6mm 1:3 mix)	240.0	m²	IS 1661:1972
Total Plaster Surface Area	Combined wall + ceiling surface	457.8	m²	IS 1200 Pt 12
Plastering Cement	Combined wall & ceiling plaster	34 (1,673 kg)	Bags 50kg	IS 1661/IS 2402
Plastering Sand	Clean plaster sand (Zone III/IV)	7.46	Tonnes	IS 383:2016
Plastering Water	W:C = 0.50	836	Litres	IS 456 Cl 5.4
Surface Primer	1 coat @ 8 m²/L	57.2	Litres	NBC 2016 SP7
Polymer Wall Putty	2 coats @ 1.5 kg/m²	1,046.7	kg	NBC 2016 SP7
Interior Acrylic Emulsion	2 coats @ 12 m²/L	58.1	Litres	NBC 2016 SP7
Exterior Weather Emulsion	2 coats @ 10 m²/L	21.8	Litres	NBC 2016 SP7
GRAND TOTAL CEMENT (Mortar+Plaster)	**All masonry + plastering combined**	**70 bags (3,471 kg)**	**Bags**	**IS 269:2015**
GRAND TOTAL SAND (Mortar+Plaster)	**All masonry + plastering sand**	**15.46**	**Tonnes**	**IS 383:2016**
GRAND TOTAL WATER (Mortar+Plaster)	**All masonry + plastering water**	**1,736**	**Litres**	**IS 456 Cl 5.4**

6. Trade-wise Cost Estimate (CPWD DSR)

Trade / Material Component	Quantity	Rate (INR)	Subtotal (INR)
Clay Modular Bricks (IS 1077)	6,264 Nos	■9 / brick	■56,376.00
Brickwork Mortar Cement	36 Bags	■380 / bag	■13,680.00
Brickwork Mortar Sand	7.99 Tonnes	■1,500 / Tonne	■11,989.83
Plastering Work (12/15/6mm)	457.8 m²	■280 / m²	■128,184.00
Surface Finishes (Primer+Putty+Paint)	457.8 m²	■120 / m²	■54,936.00
TOTAL MASONRY, PLASTER & FINISHES			**■265,165.83**

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